Year 10 Mathematics

Sets 1

9.2 Venn Diagrams

Venn Diagrams

STARTER · 5 MIN

Date: _____ **Title:** Venn Diagrams

LEARNING OBJECTIVES

- Draw a Venn diagram for two sets
- Find $A \cup B$, $A \cap B$ and A'
- Find the number of elements in a region

STARTER *Answer in your book. No calculators.*

1. List the factors of 10
2. List the even numbers from 1 to 10
3. Which numbers are in both lists?
4. Which numbers from 1 to 10 are in neither list?

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 2, 5, 10**

2. **2, 4, 6, 8, 10**

3. **2 and 10**

4. **3, 7, 9**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can draw a Venn diagram for two sets	☐
SILVER	I can find $A \cup B$, $A \cap B$ and A'	☐
GOLD	I can find the number of elements in a region	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Venn Diagrams

INTRODUCTION · 3 MIN

A Venn diagram shows sets as circles inside a rectangle ($\mathcal{E}$).

Intersection

$A \cap B$: in A AND in B.

The overlap

Union

$A \cup B$: in A OR in B (or both).

Everything inside the circles

Complement

A' : NOT in A.

Everything outside circle A

Order

Fill the overlap first.

Then the rest of each circle, then outside

Examples 1 and 2 — Drawing a Venn Diagram

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

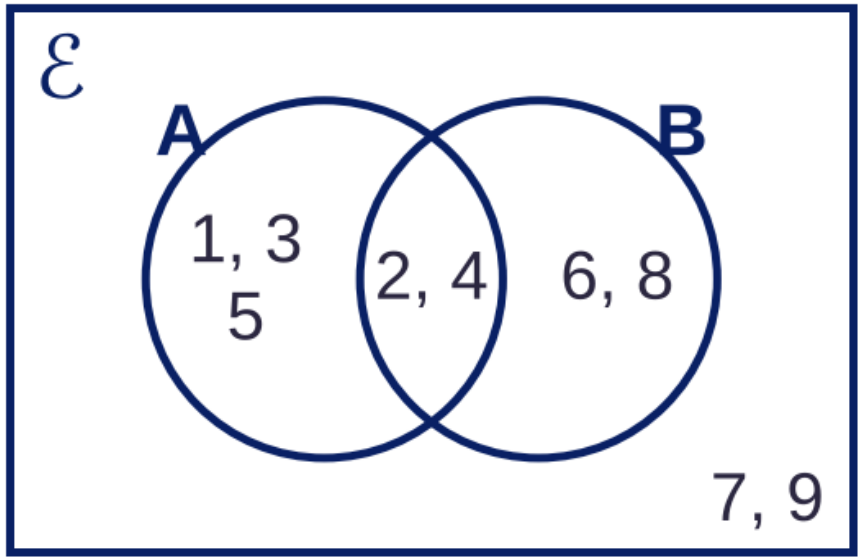
$\mathscr{E} = \{1,\ 2,\ \ldots,\ 9\}$

A = {1, 2, 3, 4, 5}, B = {2, 4, 6, 8}

Draw a Venn diagram.

Example 2

Use the Venn diagram to find n(A).



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

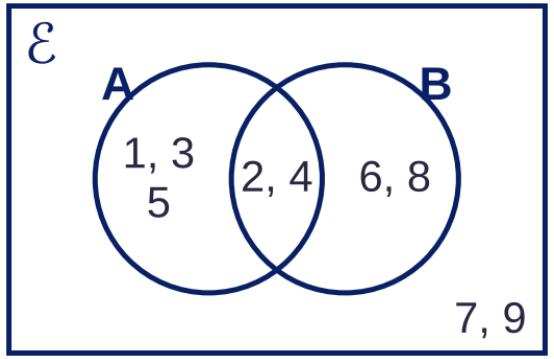
Example 1

$A \cap B = \{2, 4\}$

Only A: 1, 3, 5

Only B: 6, 8

Outside: 7, 9



Tip: put the overlap in first.

Example 2

$A = \{1, 2, 3, 4, 5\}$

$n(A) = \mathbf{5}$

Tip: count everything inside circle A.

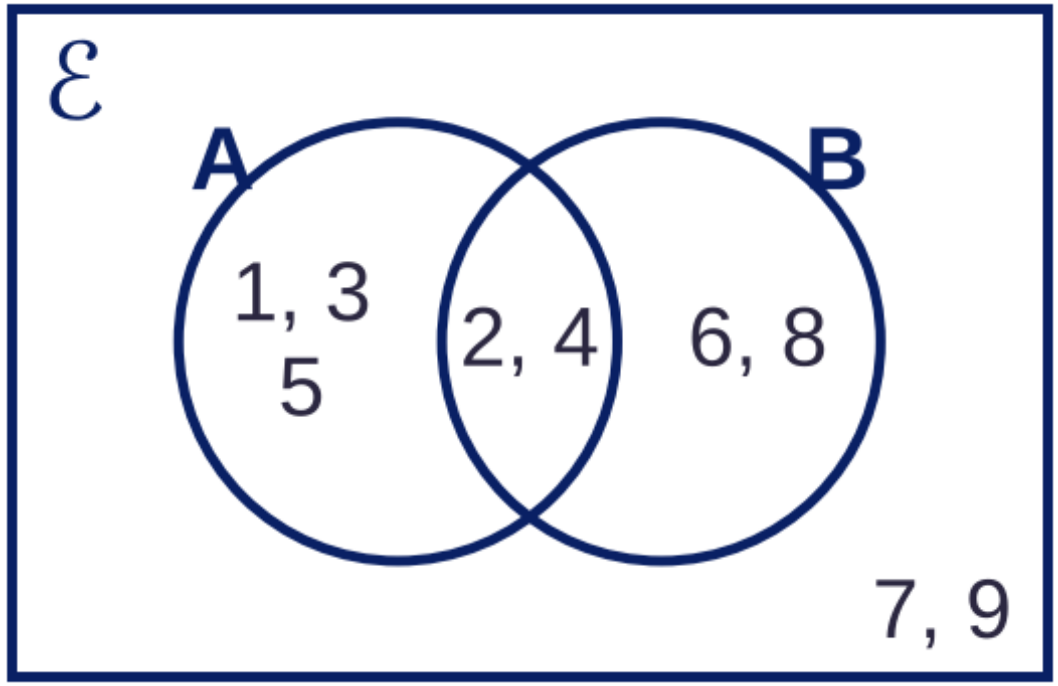
Task 1

PRACTICE · 9 MIN

BRONZE *Use the Venn diagram*

1. List $A \cap B$
2. List the elements of B
3. List $A \cup B$
4. List A'
5. Find $n(A \cup B)$

★ List the elements in neither A nor B



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $A \cap B = \{2, 4\}$
2. $B = \{2, 4, 6, 8\}$
3. $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$
4. $A' = \{6, 7, 8, 9\}$
5. $n(A \cup B) = 7$

★ $\{7, 9\}$

Examples 3 and 4 — Union, Intersection and Complement

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

$\mathscr{E} = \{1,\ 2,\ \ldots,\ 12\}$

P = {multiples of 3},
Q = {factors of 12}
Find $P \cap Q$ and $P \cup Q$.

Example 4

Using the sets in Example 3,

find $(P \cup Q)'$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$P = \{3, 6, 9, 12\}$

$Q = \{1, 2, 3, 4, 6, 12\}$

$P \cap Q = \{3, 6, 12\}$

$P \cup Q$ **$= \{1, 2, 3, 4, 6, 9, 12\}$**

Tip: ∩ is the overlap, ∪ is everything in either.

Example 4

$P \cup Q = \{1, 2, 3, 4, 6, 9, 12\}$

$(P \cup Q)'$ **$= \{5, 7, 8, 10, 11\}$**

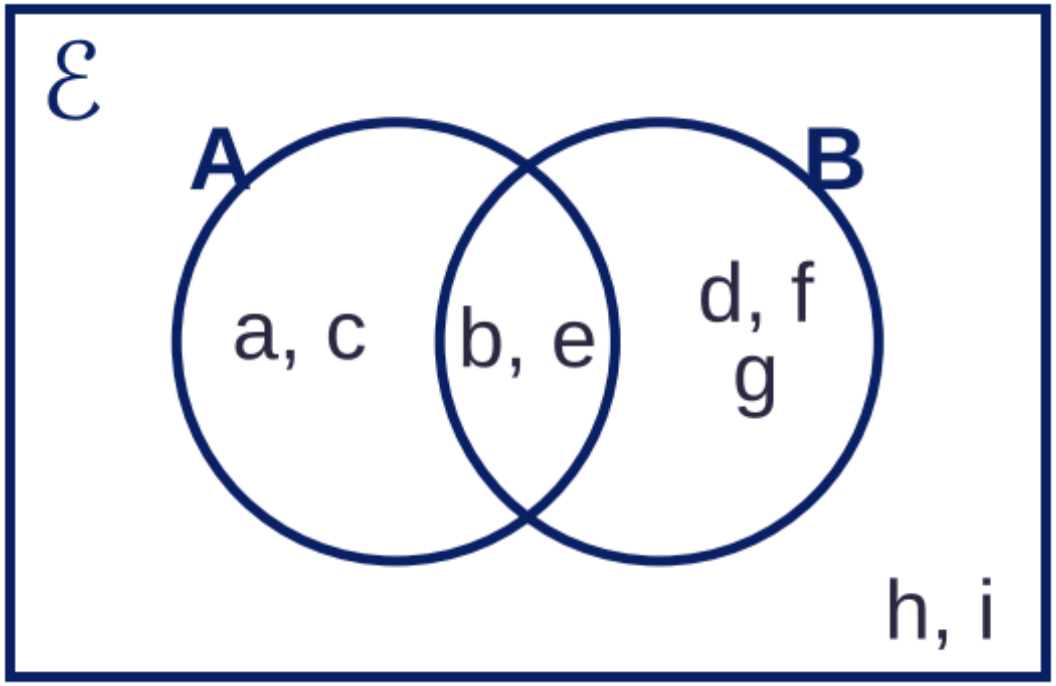
Tip: ′ means 'not in'.

Task 2

PRACTICE · 9 MIN

SILVER *Use the Venn diagram*

1. List $A \cap B$
2. List $A \cup B$
3. List B'
4. Find $n(A')$
5. List $(A \cup B)'$



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $A \cap B = \{b, e\}$

2. $A \cup B = \{a, b, c, d, e, f, g\}$

3. $B' = \{a, c, h, i\}$

4. $A' = \{d, f, g, h, i\}, \quad n(A') = 5$

5. $(A \cup B)' = \{h, i\}$

Examples 5 and 6 — Numbers in Regions

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5

(2 marks)

$\mathscr{E} = \{1,\ 2,\ \ldots,\ 15\}$

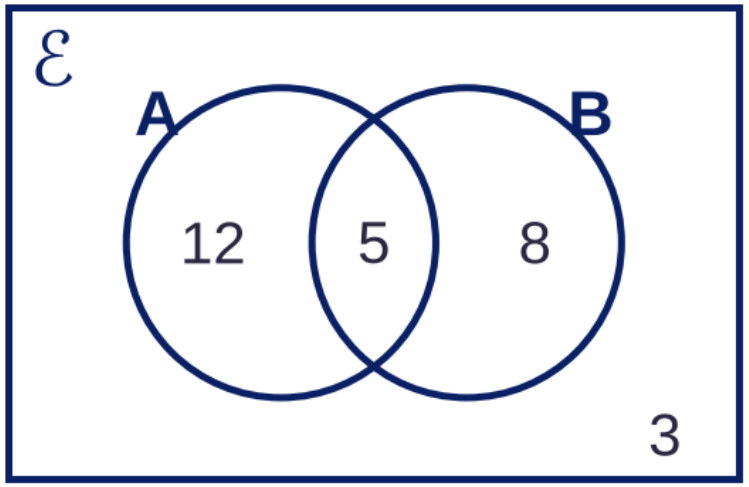
E = {even numbers},
F = {multiples of 3}

Find $n(E \cap F)$ and $n(E \cup F)$.

Example 6

(2 marks)

The Venn diagram shows the number of elements in each region. Find n(A).



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$E \cap F = \{6, 12\}$

$n(E \cap F) = 2$

$E \cup F = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15\}$

$n(E \cup F) = \mathbf{10}$

Tip: list the sets first.

Example 6

$n(A) = 12 + 5$

$n(A) = \mathbf{17}$

Tip: the numbers are counts, not elements.

Task 3

PRACTICE · 9 MIN

GOLD

The Venn diagram shows the number of elements in each region

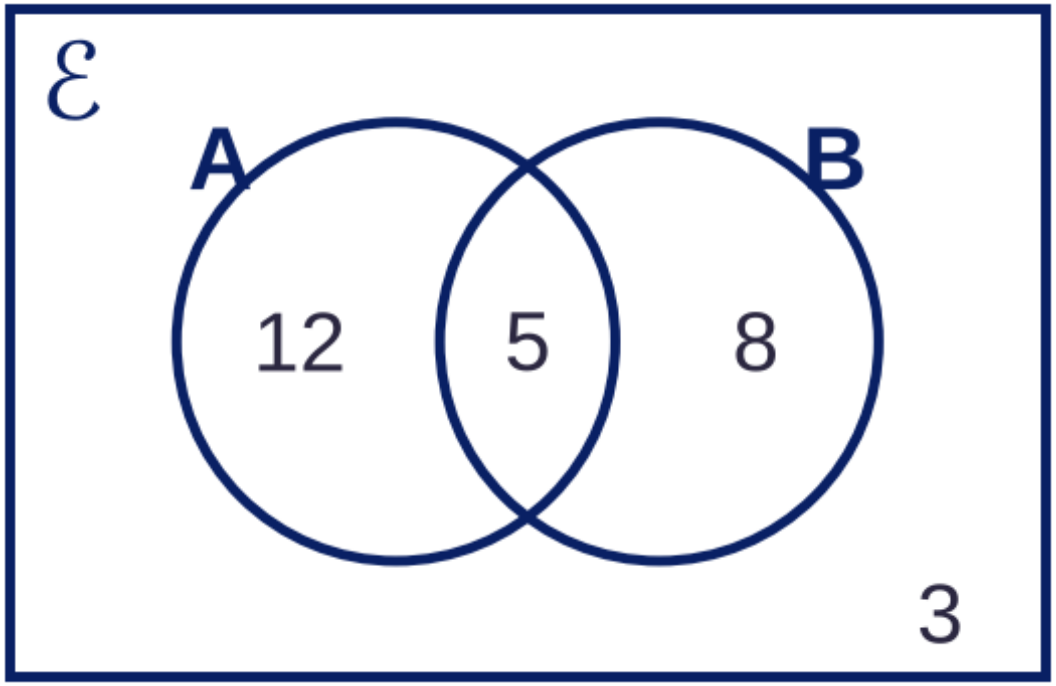
1. Find n(B) (1 mark)

2. Find $n(A \cap B)$ (1 mark)

3. Find $n(A \cup B)$ (1 mark)

4. Find $n(\mathscr{E})$ (1 mark)

5. Find $n(A')$ (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $n(B) = 5 + 8 = \mathbf{13}$
2. $n(A \cap B) = \mathbf{5}$
3. $n(A \cup B) = 12 + 5 + 8 = \mathbf{25}$
4. $n(\mathscr{E}) = 12 + 5 + 8 + 3 = \mathbf{28}$
5. $n(A') = 8 + 3 = \mathbf{11}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $\mathscr{E} = \{1,\ \ldots,\ 8\},\ A = \{1,\ 2,\ 3,\ 4\},\ B = \{3,\ 4,\ 5,\ 6\}.$ List $A \cap B$.

SILVER Using the same sets, list $A \cup B$.

GOLD Using the same sets, list $(A \cup B)'$.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $A \cap B = \{3,\ 4\}$

SILVER $A \cup B = \{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$

GOLD $(A \cup B)' = \{7,\ 8\}$

Exam Practice — Drawing Venn Diagrams

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

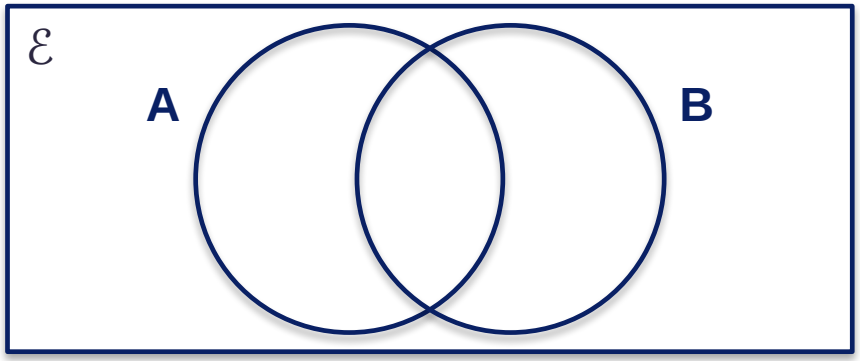
(2 marks)

$\mathcal{E}$ = {1, 2, 3, …, 10}

A = {factors of 10}

B = {even numbers}

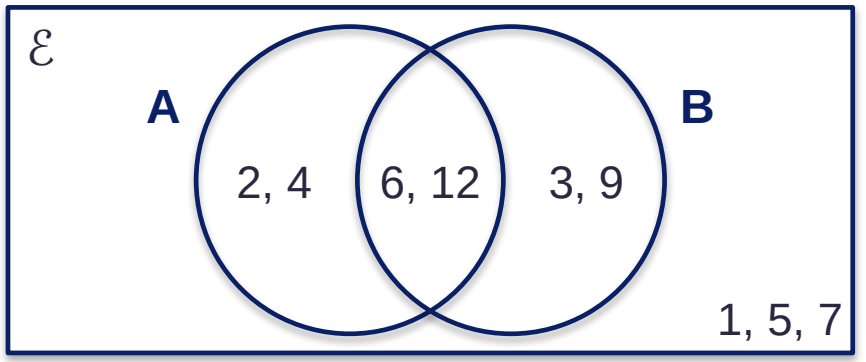
Complete the Venn diagram.



Question 2

(2 marks)

Here is a Venn diagram.



(a) List the elements of A ∩ B. (1)

.....

(b) Find n(B). (1)

n(B) = ...

Exam Practice — Union and Complement

EXAM PRACTICE · 10 MIN

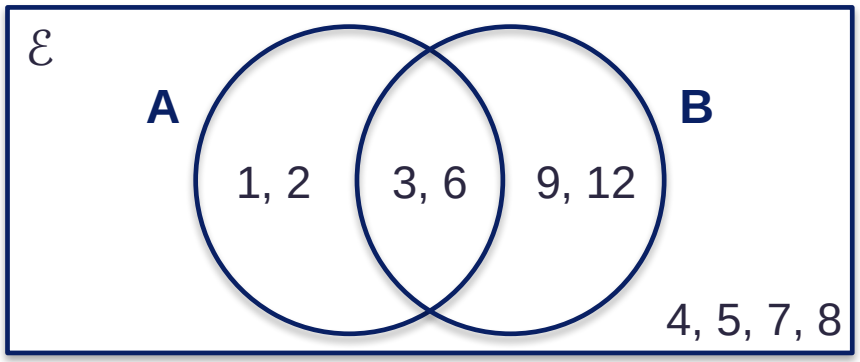
SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

Here is a Venn diagram.



(a) List the elements of A ∪ B. (1)

.....

(b) List the elements of A′. (1)

.....

Question 4

(2 marks)

ℰ = {1, 2, 3, …, 15}

P = {multiples of 3}

Q = {odd numbers}

(a) List the elements of P ∩ Q. (1)

.....

(b) Find n(P ∪ Q). (1)

n(P ∪ Q) = ...

Exam Practice — Numbers in Regions

EXAM PRACTICE · 10 MIN

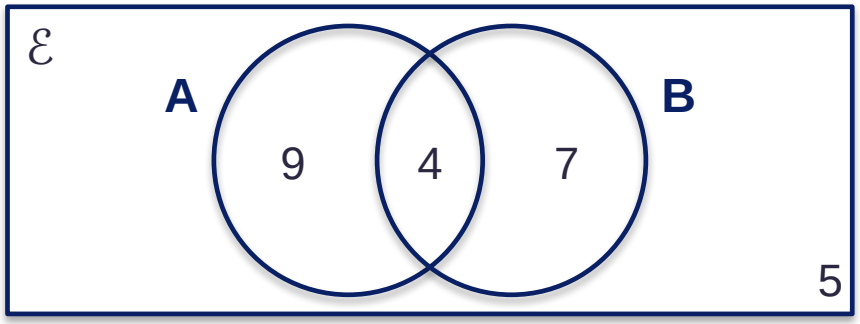
GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

The Venn diagram shows the number of elements in each region.



(a) Find n(A ∪ B). (1)

n(A ∪ B) = ...

(b) Find n(A′). (1)

n(A′) = ...

Question 6

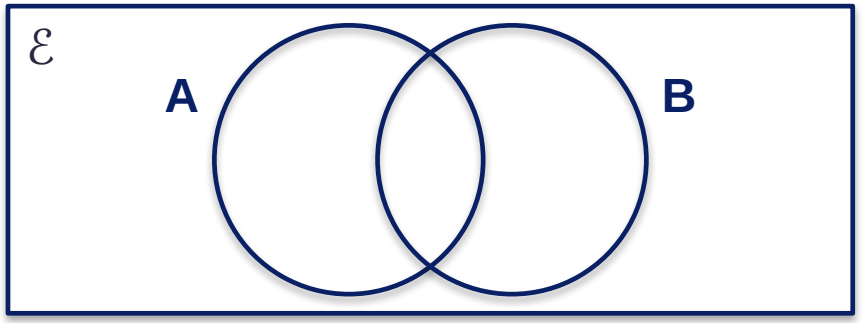
(3 marks)

n(ℰ) = 40, n(A) = 22

n(B) = 15, n(A ∩ B) = 6

Find n((A ∪ B)′).

You may use the Venn diagram.



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Exam Practice — Mark Scheme

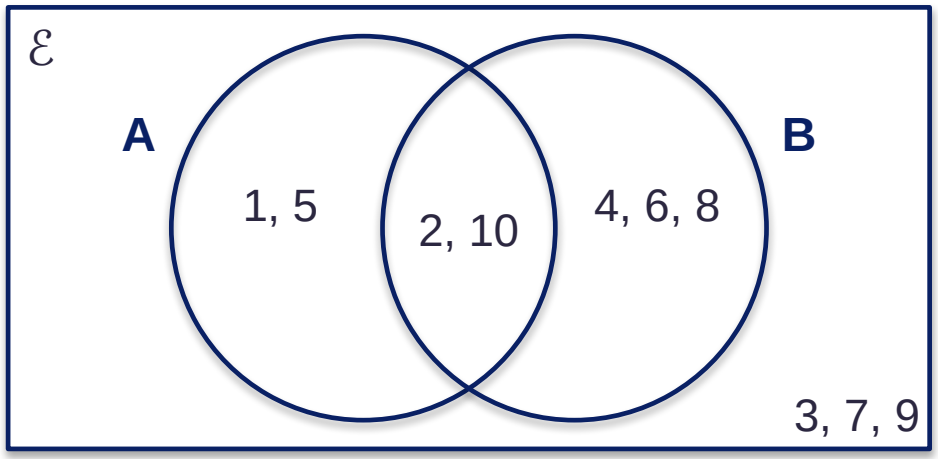
CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

All 10 numbers correct B2



Tip: B1 for at least 7 numbers correct.

Question 2

(a) {6, 12} B1

(b) n(B) = 4 B1

Tip: n(B) counts all of circle B.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) {1, 2, 3, 6, 9, 12} B1

(b) {4, 5, 7, 8, 9, 12} B1

Tip: A′ is everything outside circle A.

Question 4

(a) {3, 9, 15} B1

(b) {1, 3, 5, 6, 7, 9,

11, 12, 13, 15}

n(P ∪ Q) = 10 B1

Tip: count 3, 9 and 15 only once.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 9 + 4 + 7 = 20 — B1

(b) 7 + 5 = 12 — B1

Tip: the numbers are counts, not elements.

Question 6

22 − 6 = 16 and 15 − 6 = 9 — M1

40 − (16 + 6 + 9) — M1

= 9 — A1

Tip: put n(A ∩ B) = 6 in the overlap first.